

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. We distinguish the deterministic flow, Koopman observables, probability laws, factorial hierarchies, and limiting kinetic values. The dynamic entropy action has compact sublevels and defines a strongly continuous Nisio semigroup. Its discounted variational resolvent satisfies the nonlinear resolvent identity, has full range, and generates an m -dissipative graph with comparison. State-dependent controls are transferred by their relative multiplier rather than by reusing one collision measure. A single diagonal BBGKY graph-core corrector verifies all nonlinear Trotter–Kato hypotheses.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: STATE-DEPENDENT CONTROLS, NONLINEAR RESOLVENTS, AND KINETIC SEMIGROUP CONVERGENCE

The finite deterministic gas, its probability law, its factorial hierarchy, and the limiting density value are kept as distinct typed objects. The kinetic transition semigroup is constructed from the dynamic B2 action; no initial preparation cost is charged inside a transition. Comparison is obtained from

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the nonlinear resolvent and a state-dependent control-transfer lemma, not by applying one collision measure to two different density paths.

1.1. Exact finite objects and the law–hierarchy chart. Let Φ_t^ε be the hard-sphere flow on the regular phase space, $U_t^\varepsilon F = F \circ \Phi_t^\varepsilon$ its Koopman group, and $(\Phi_t^\varepsilon)_\# P$ the push-forward of an initial law. On laws with a uniform exponential particle-number and energy moment, let $\mathfrak{K}(P) = (g_j(P))_{j \geq 0}$ be the factorial-correlation transform. Equip the hierarchy with

$$\|g\|_{\alpha, \beta} = \sum_{j \geq 0} \frac{e^{\alpha j}}{j!} \int |g_j(Z_j)| e^{\beta |V_j|^2} dZ_j.$$

Theorem 1.1 (Exact law–hierarchy realization). *On the exponential-moment law class, $\mathfrak{K}$ is injective, its image is characterized by Lenard positivity, consistency, and the exponential bound, and*

$$\mathfrak{K}((\Phi_t^\varepsilon)_\# P) = \mathcal{U}_t^\varepsilon \mathfrak{K}(P),$$

where $\mathcal{U}_t^\varepsilon$ is the BBGKY hierarchy evolution on the realizable cone. Polynomial law cylinders and their exponential transforms are determined by absolutely convergent factorial series on this cone.

Proof. The exponential particle-number bound makes the Janossy generating functional entire near the unit function. Its derivatives are the correlation functions, so equality of all g_j implies equality of the generating functional and hence of the law. Lenard positivity and consistency reconstruct the finite-volume Janossy measures; the exponential bound normalizes their sum. Integrating the Liouville equation against factorial observables gives BBGKY and proves the intertwining. The factor $1/j!$ together with the exponential moment gives normal convergence of polynomial and exponential cylinders. $\square$

1.2. The dynamic action and state-dependent control transfer. For a positive density path f and a contact intensity $d\Gamma = q dA_f$, define the transition action

$$\mathcal{A}_{s,t}(f, \Gamma) = \int_s^t \int \ell(q) dA_f, \quad \ell(q) = q \log q - q + 1,$$

subject to the exact weak balance and the initial condition $f(s) = f_s$. The initial preparation rate is absent.

Lemma 1.2 (Multiplier transfer between nearby states). *Let $(f, \Gamma = qA_f)$ be a regular controlled path with $0 < q_- \leq q \leq q_+$, and let g_s be close to f_s in the weighted bounded-Lipschitz metric and in energy. Then there is a balanced path $(g, \tilde{\Gamma})$ beginning at g_s with*

$$\tilde{\Gamma} = qA_g + R(f, g),$$

where R is an exactly conservative correction satisfying

$$\|R(f, g)\|_{L_w^1} + |\mathcal{A}(g, \tilde{\Gamma}) - \mathcal{A}(f, \Gamma)| \leq C d_*(f_s, g_s).$$

The correction preserves strict positivity for sufficiently close states.

Proof. Use the same relative multiplier q , not the same collision measure. The biased Boltzmann equation with intensity qA_g is locally well posed in the regular weighted chart, and stability of its mild equation gives $\sup_t d_*(f_t, g_t) \leq Cd_*(f_s, g_s)$. The residual in the finite family of mass, momentum, energy, and balance coordinates is therefore $O(d_*)$. Apply the bounded right inverse for the linearized balance from Paper B3 to a smooth space–time approximation of that residual, and add a fixed positive Maxwellian collision background of size $2Cd_*$. The resulting correction has negative part dominated by the background and hence preserves positivity. The perspective entropy is locally Lipschitz on $q \in [q_-/2, 2q_+]$; its weighted integral gives the action estimate. $\square$

Lemma 1.3 (Compact dynamic action sublevels). *Fix a compact initial family with uniformly bounded mass and energy. For each M , the set of balanced paths satisfying $\mathcal{A}_{0,T}(f, \Gamma) \leq M$ is compact in the topology of weighted weak density paths and weak contact measures, strengthened by the time modulus on a countable smooth test family.*

Proof. The entropy inequality $qr \leq \ell(q) + e^r - 1$ controls the contact mass against every bounded weighted test. Elastic balance conserves mass, momentum, and energy, while a superlinear velocity test and the fixed initial exponential moment give uniform velocity tightness. The weak balance bounds increments of $\langle f_t, \phi_k \rangle$ by transport moments plus contact mass, yielding a uniform modulus. Prokhorov, Arzelà–Ascoli on the determining coordinates, and closedness of the balance relation give sequential compactness. Lower semicontinuity follows from the joint lower semicontinuity of the perspective integrand $(f, \Gamma) \mapsto \int d\Gamma \log(d\Gamma/dA_f) - d\Gamma + dA_f$. $\square$

1.3. Nisio semigroup and nonlinear resolvent. For $h \in \text{BUC}(\mathcal{E})$ define

$$(S_t h)(f_0) = \sup_{(f, \Gamma): f(0)=f_0} \{h(f_t) - \mathcal{A}_{0,t}(f, \Gamma)\}.$$

Theorem 1.4 (Kinetic Nisio semigroup). *The operators $(S_t)_{t \geq 0}$ form a strongly continuous, monotone, cash-additive contraction semigroup on weighted uniformly continuous functions. For every $\lambda > 0$ the nonlinear resolvent*

$$R_\lambda h = \sup_{(f, \Gamma): f(0)=f_0} \int_0^\infty e^{-\lambda t} [\lambda h(f_t) - \ell(d\Gamma/dA_f) A_f(dt)]$$

is well defined, satisfies

$$R_\lambda - R_\mu = (\mu - \lambda) R_\lambda R_\mu$$

in the nonlinear resolvent sense, and is a sup-norm contraction.

Proof. Concatenation of balanced paths preserves the state at the joining time and adds the dynamic action, proving the semigroup law in both directions. Lemma 1.3 gives attainment after relaxation. Strong continuity follows from

the short-time balance estimate and the zero-cost Boltzmann path. For the resolvent, split the discounted integral at a stopping time s , optimize the tail conditional on f_s , and use the semigroup law. Integrating the resulting dynamic-programming identity against $\lambda e^{-\lambda s}$ gives the nonlinear resolvent identity. Taking the same admissible path in two terminal payoffs gives contraction; no control is transported between distinct initial states in this argument. $\square$

Theorem 1.5 (Full range, m -dissipativity, and comparison). *Define the graph $\mathcal{H}$ by*

$$(u, g) \in \mathcal{H} \iff u = R_\lambda(u + \lambda^{-1}g)$$

for one, hence every, $\lambda > 0$. Then $\mathcal{H}$ is closed, m -dissipative, and independent of λ . The range of $I - \lambda^{-1}\mathcal{H}$ is all of $\text{BUC}(\mathcal{E})$. Consequently the resolvent equation and the evolution equation have comparison, and S_t is the unique nonlinear semigroup generated by $\mathcal{H}$.

Proof. The resolvent identity in Theorem 1.4 proves independence of λ and the range statement directly: for every h , $u = R_\lambda h$ solves $h \in u - \lambda^{-1}\mathcal{H}u$. Contraction gives dissipativity and closedness. The nonlinear Hille–Yosida–Crandall–Liggett theorem then produces a unique contraction semigroup. Its exponential formula agrees with S_t because both satisfy the same dynamic-programming resolvent. Comparison is the contraction estimate for sub- and supersolutions of the resolvent, followed by the exponential formula. Lemma 1.2 supplies continuity in the state needed to pass from the regular core to the closure. $\square$

1.4. Observable graph core and a single perturbed-test sequence.

Let $\mathcal{D}_\varepsilon$ be the algebra of smooth symmetric observables satisfying specular boundary compatibility, completed in the graph norm of the Liouville generator. Let $\mathcal{D}$ be the cylinder core on densities whose collision increment is bounded.

Theorem 1.6 (Graph-core hierarchy corrector). *For every $F \in \mathcal{D}$ there are integers $K(\varepsilon) \uparrow \infty$ and exact observables $F_\varepsilon \in \mathcal{D}_\varepsilon$ such that*

$$F_\varepsilon \rightarrow F, \quad \mathcal{H}_\varepsilon F_\varepsilon \rightarrow \mathbb{H}\left(f, \frac{\delta F}{\delta f}\right)$$

locally uniformly on action sublevels. The construction uses the full triangular BBGKY Duhamel chain with $1/j!$ weights and yields one diagonal sequence, not an iterated limit.

Proof. Galerkin-project the observable generator to the first K hierarchy levels and solve the finite terminal corrector equation by the forward remaining-time propagator. The j th connected defect is bounded by the B2 factorial majorant $j!C^jT^{j-1}$; multiplication by the hierarchy weight $e^{\alpha j}/j!$ leaves a geometric tail $C\rho^K$. The finite- ε recollision defect is bounded by $C_K\varepsilon^\gamma$. Choose $K(\varepsilon)$ increasing slowly enough that $\rho^{K(\varepsilon)} + C_{K(\varepsilon)}\varepsilon^\gamma \rightarrow 0$. Resolvent regularization in the Liouville graph norm puts every finite corrector in $\mathcal{D}_\varepsilon$

while preserving the boundary jump. The ensemble boundary identity then gives the exponential collision factor $e^{\Delta p} - 1$; it is not attributed to the deterministic Koopman chain rule. $\square$

Theorem 1.7 (Nonlinear Trotter–Kato convergence). *The finite ensemble log-Laplace semigroups converge locally uniformly on compact action sublevels to S_t . The hypotheses of nonlinear Trotter–Kato hold: compact containment, a separating graph core, upper and lower generator recovery, and uniqueness of the limiting resolvent.*

Proof. B2 gives exponential compact containment and the joint dynamic action. The corrector theorem supplies both generator inequalities on a separating core. Lemma 1.3 gives equicoercivity. Half-relaxed resolvent limits are respectively sub- and supersolutions of the limiting graph; the comparison theorem identifies them with R_λ . Nonlinear Trotter–Kato therefore gives resolvent and semigroup convergence. Exact law–hierarchy realizability from Theorem 1.1 ensures that the microscopic values are computed from genuine probability laws, not from arbitrary formal hierarchies. $\square$

Theorem 1.8 (Round-seventeen B4 closure). *The hard-sphere ensemble admits a typed exact law–hierarchy representation, a compact dynamic action, a strongly continuous kinetic Nisio semigroup, an m -dissipative nonlinear resolvent with comparison, a single graph-domain corrector sequence, and microscopic nonlinear-semigroup convergence.*

Proof. Combine Theorems 1.1, 1.4, 1.5, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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